



OPEN SleepPACNet: new convolutional neural network considering phase-amplitude coupling for automatic sleep stage classification using single-channel electroencephalogram

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Automatic sleep stage classification (ASSC) using single-channel electroencephalogram (EEG) signals is an essential technology for developing wearable EEG devices for sleep monitoring. Deep learning (DL)-based methods, although effective in automatically extracting features, often overlook significant EEG characteristics containing domain-specific knowledge relevant to particular sleep stages, such as cross-frequency phase-amplitude coupling (PAC) phenomena. To address this limitation, we proposed SleepPACNet, a novel convolutional neural network (CNN) architecture specifically designed to effectively extract PAC-related features from single-channel EEG signals to enhance ASSC performance. Central to our proposed model is a PAC feature extraction module, which employs a CNN layer explicitly designed to integrate the phase information from low-frequency EEG components, obtained using the Hilbert transform, with amplitude information from high-frequency EEG components. The performance of SleepPACNet was evaluated using single-channel (Fp1–Fp2) sleep EEG data extracted from the Dreem dataset to simulate conditions of a portable EEG device measuring prefrontal bipolar EEG signals. SleepPACNet outperformed conventional CNN models, including SleepCNN, achieving the highest classification accuracy of 75.7%. It significantly improved REM stage classification, validating PAC's effectiveness in REM detection. These findings highlight the potential of automated PAC feature extraction in DL for improving the overall performance of ASSC systems utilizing single-channel EEG.

Keywords Automatic sleep stage classification (ASSC), Electroencephalography (EEG), Phase-amplitude coupling, Hilbert transform, Convolutional neural network (CNN)

Sleep is an important human activity for maintaining both physical and mental health in humans^{1,2}. During sleep, the stages of sleep change according to the brain's state³. Sleep stages are generally classified into the following five stages: wake (W), rapid eyes movements (REM), and three non-REM stages (referred to as N1, N2, and N3)⁴. Identifying these sleep stages is essential for the diagnosis of sleep disorders and measurement of sleep quality^{5,6}. Sleep experts typically perform manual scoring of the sleep stages based on established guidelines, such as the Rechtschaffen and Kales (R&K) and the American Academy of Sleep Medicine (AASM), typically using polysomnography (PSG), which encompasses various biomedical signals, including electroencephalography (EEG), electromyography (EMG), and electrooculography (EOG)^{7,8}. However, this manual scoring method requires considerable effort and time investment from sleep experts, and the excessive workload can sometimes reduce the accuracy and reliability of sleep stage scoring^{9,10}. In addition, attaching a large number of sensors and sleeping in an unfamiliar environment can cause discomfort for participants, and the measurement procedure

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itself may disrupt sleep¹¹. To address these issues, automatic sleep stage classification (ASSC) using fewer sensors has been investigated. The advancement of such techniques is anticipated to facilitate the development of wearable devices capable of conveniently diagnosing sleep disorders or assessing sleep quality in home environments.

Among the biomedical signals utilized in PSG, EEG is considered the most critical modality for sleep-stage scoring due to its rich neurophysiological information¹². This characteristic has prompted numerous ASSC studies to focus exclusively on EEG signals. Specifically, since scalp electrodes attached to hairy area may interfere with natural sleep¹³, ASSC based on traditional machine learning (ML) methods using single-channel prefrontal EEG recordings has attracted considerable attention as a means to minimize the number of required electrodes. For instance, Liang et al.¹⁴ employed linear discriminant analysis (LDA) to classify sleep stages by extracting handcrafted features, including autoregressive coefficients and multiscale entropy. Handcrafted features used for traditional ML-based ASSC are advantageous because they can explicitly incorporate domain knowledge¹⁵, thereby making the classification process interpretable¹⁶. However, the use of handcrafted features may limit classification performance due to potential loss of information necessary for classification¹⁷.

Recently, many studies have demonstrated the potential of deep learning (DL)-based ASSC using single-channel EEG, which can automatically extract features from raw EEG signals with minimal information loss. Particularly, most DL-based ASSC studies have employed convolutional neural networks (CNNs), which showed exceptional performance in feature extraction from EEG. For instance, Goshtasbi et al.¹⁸ introduced a CNN architecture that integrates a feature extraction module with a temporal sequence learning module. Mousavi et al.¹⁹ developed the SleepEEGNet, consisting of CNN for extracting features from time-frequency information of EEG and a sequence-to-sequence model for recognizing contextual dependencies between continuous EEG signals. Additionally, the TinySleepNet, proposed by Supratak et al.²⁰, also includes a CNN-based feature extraction module. However, while CNN-based feature extractors effectively capture patterns from raw EEG data, they often underperform due to limited generalization and insufficient domain knowledge. With recent advancements in large language models, large-scale foundation models for EEG, such as EEGPT²¹, NeuroGPT²², LaBram²³, and BCINetV1²⁴, have been developed to improve generalization through pre-training on extensive datasets. Alongside these foundation models, recent multidomain frameworks have also been proposed to achieve cross-discipline adaptability and robust generalization²⁵. Despite their enhanced generalization, these models may not fully leverage some domain knowledge that was successfully embedded in conventional handcrafted features^{15,16}, highlighting the need for further model development in this regard. Therefore, incorporation of specific domain knowledge into DL-based ASSC has the potential to increase classification performance, as shown in previous DL-based EEG studies that attempted to incorporate handcrafted features into DL in other EEG classification problems^{26–28}.

Among the diverse handcrafted EEG features employed for traditional ML-based ASSC using single-channel EEG, cross-frequency phase-amplitude coupling (PAC) has shown potential as an effective EEG feature for ASSC. PAC is typically quantified using diverse metrics, such as envelope-to-signal (ESC), Modulation Index (MI), and cross-frequency coherence (CFC)^{29–31}. Specifically, ESC measures the correlation between the low-frequency signal and the high-frequency envelope, while MI is calculated as the mean of a composite complex signal formed by the low-frequency phase and high-frequency amplitude envelope. Additionally, CFC quantifies the coherence between the high-frequency energy and the raw low-frequency signal. In the physiological aspect, the phase of the delta wave in the prefrontal cortex synchronizes with the amplitude of the spindle wave in the thalamus during sleep^{32,33}. Manasova et al.³⁴ found that delta-alpha PAC, which implies synchronization of delta wave phase and sleep spindle wave amplitude, significantly increases in the N2 and N3 stages compared to other sleep stages. Based on this phenomenon, Dimitriadis et al.³⁴ employed the delta-alpha PAC, calculated using various metrics including MI, as one of the key features for ML-based ASSC, applying traditional classifiers such as k-nearest neighbor, the Naïve Bayesian, linear discriminant analysis, and support vector machines to achieve enhanced classification performance.

Given the considerable potential of PAC, incorporating PAC information into DL is expected to enhance the overall performance of ASSC. Nevertheless, PAC presents challenges in its direct application to DL-based classification. First, using calculated PAC features directly for DL-based classification may yield varying performances due to the diverse methods to calculate the PAC measures³⁶. Second, the process of computing PAC features may result in information loss, potentially omitting critical characteristics of the original raw EEG signals. To address these problems, Formoso et al. recently employed a parallel frequency-band DL architecture to automatically extract the PAC-related information without human intervention; however, this approach implicitly provides the network with phase information in a low-frequency signal. To more effectively capture PAC-related information, it is necessary to explicitly represent the low-frequency phase, thereby facilitating the utilization of phase information by the DL model.

In this study, we attempted to automatically extract PAC features using specially designed DL architecture incorporating the Hilbert transform. Specifically, we propose SleepPACNet, a novel CNN-based DL architecture for ASSC, which was designed to effectively extract PAC features by leveraging phase information revealed through the Hilbert transform from a prefrontal single-channel EEG signal. SleepPACNet consists of two parallel components: the conventional feature extraction module and PAC feature extraction module. To enable CNN in SleepPACNet to capture the PAC information, we employed Hilbert transform to reveal the phase information in low-frequency EEG signal³⁷, and the low-frequency phase information was integrated with the high-frequency amplitude information. To validate the classification performance of SleepPACNet, a bipolar prefrontal single-channel EEG signal was generated from the Dreem Open Datasets (DOD)³⁸. Furthermore, to show the potential contribution of the PAC feature extraction module in SleepPACNet in classifying sleep stages, we compared the classification performance of the SleepPACNet with that of the CNN model containing only the conventional feature extraction module as well as those of state-of-the-art DL models for ASSC. To the best

of our knowledge, SleepPACNet is the first DL model developed to utilize phase information extracted via the Hilbert transform for the automatic extraction of PAC features from EEG.

Overall, the contributions of our study are as follows:

- We propose a novel DL architecture in which Hilbert-transformed low-frequency EEG signal and high-frequency EEG signal jointly pass through CNN layers to extract features that incorporate PAC information, under the assumption that the Hilbert-transformed EEG signal includes phase information, while the raw EEG signal contains amplitude information. This extraction strategy facilitates the extraction of features embedding meaningful domain knowledge while minimizing information loss, thereby addressing the limitations of traditional ML and DL approaches.
- Our results suggest the possibility that our model could be utilized with a home-based portable EEG device, as demonstrated by the validation using a bipolar prefrontal single-channel EEG signal for sleep stage classification. Through the extensive analyses, we demonstrate that our model outperformed state-of-the-art models for ASSC as well as our model without the PAC feature extraction module.

Materials and methods

This section first provides a comprehensive overview of the experimental data utilized to assess the classification performance of DL models. Then, it presents the detailed architecture of SleepPACNet, including the function of each component, as depicted in Fig. 1. The detailed architectural specifications and hyperparameters of SleepPACNet are provided in Supplementary information to ensure the reproducibility of the study (Supplementary Table 1).

Experimental data

Previous research^{32,33} has demonstrated that single-channel EEG signals recorded from the prefrontal region can benefit significantly from incorporating PAC features. Additionally, for implementing portable EEG devices intended for home use, bipolar prefrontal EEG channels obtained from the forehead region are typically preferred due to lower noise levels and greater ease of measurement. Consequently, we aimed to utilize a bipolar prefrontal EEG signal recorded via a single bipolar channel (Fp1–Fp2). However, most publicly available EEG databases do not contain such bipolar EEG recordings. Although the DOD³⁸ also lacks direct Fp1–Fp2 signals, it provides prefrontal and frontal EEG channels such as Fp1–F3, Fp2–F4, and F3–F4, as detailed in the Supplementary information. These available channels allow simulation of bipolar prefrontal EEG signals suitable for portable EEG devices. Accordingly, we generated a bipolar prefrontal EEG signal, Fp1–Fp2, by combining Fp1–F3, Fp2–F4, and F3–F4 channels, as follows:

$$Fp1 - Fp2 = Fp1 - F3 - Fp2 - F4 + F3 - F4 \quad (1)$$

The PSG data in the DOD were recorded at a sampling rate of 250 Hz from 25 healthy participants. Sleep stages were manually scored into Wake (W), N1, N2, N3, REM, and ‘Not-scored’ at 30-s intervals by trained experts in accordance with the AASM scoring guidelines. The transformed Fp1–Fp2 EEG signals were segmented into non-overlapping 30-s intervals to correspond with the identified sleep stages, with segments classified as ‘Not-scored’ being excluded from subsequent analyses. Each segment was then down-sampled to 100 Hz prior to z-score normalization to enhance overall computational efficiency.

Architecture of proposed model

SleepPACNet is composed of two parallel modules: the PAC feature extraction module and conventional raw EEG feature extraction module. The PAC feature extraction module captures PAC information by leveraging the phase of low-frequency EEG signals in conjunction with the amplitude of high-frequency EEG signals. In contrast, the raw EEG feature extraction module is designed to extract both temporal and frequency-domain information from unprocessed EEG signals.

1. PAC feature extraction module

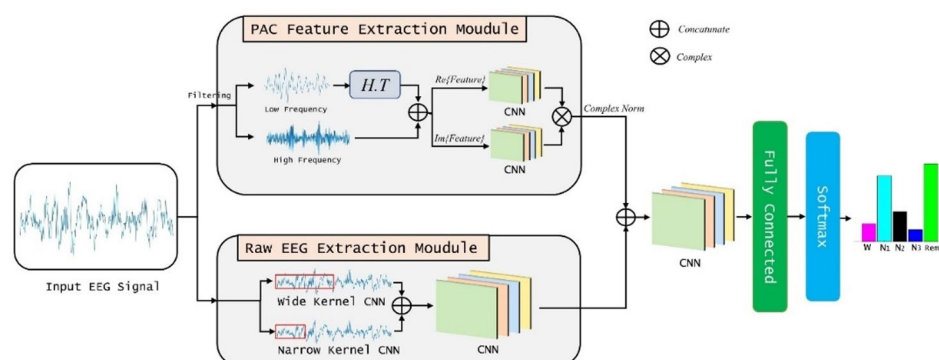


Fig. 1. The overall architecture of SleepPACNet.

To extract PAC features from EEG signals, it is necessary to use both the amplitude information, which is inherently present in the raw EEG signal, and the phase information, which is not explicitly observable. In this paper, to extract the phase information from the low-frequency EEG signal, raw EEG signal was bandpass filtered using a 6th-order zero-phase Butterworth infinite impulse response (IIR) filter with passband of 0.5–8 Hz, and then Hilbert transform was applied to the filtered EEG data. The high-frequency EEG signal was obtained by applying a 6th-order zero-phase Butterworth IIR filter with passband of 8–30 Hz. These frequency bands were selected to take into account the frequency ranges for delta waves (0.5–4 Hz) and spindle waves (12–15 Hz). These two inputs were concatenated along the feature axis, allowing the CNN to explore the interaction between the phase of the low-frequency signal and the amplitude of the high-frequency signal. However, direct feature concatenation of the two inputs was not feasible, as the low-frequency signal was a complex value owing to the Hilbert transform. Therefore, feature concatenation was performed on the high-frequency signal with both the real part and imaginary part of the Hilbert-transformed low-frequency signal. As a result, two CNN input features were obtained: F_r , representing the real part, F_i , representing the imaginary part, where $F_r = (f_1, f_2, \dots, f_T) \in \mathbb{R}^{2 \times T}$ and $F_i = (f_1, f_2, \dots, f_T) \in \mathbb{R}^{2 \times T}$, where T represents the number of time points.

Figure 2 illustrates the detailed architecture used to extract PAC features from the obtained features F_r and F_i . To investigate the relationship between the phase of low-frequency components and the amplitude of high-frequency components by integrating them, three procedures were conducted on the features F_r and F_i . First, a 2D CNN layer with a kernel size of (2×1) and stride size of (1×1) was employed for each feature to capture the simultaneous relationship between the phase of low frequency EEG and the amplitude of high frequency EEG. Second, the following 2D CNN layer with kernel size of 100, to account for the PAC cycle locked to the delta wave, was applied to each feature map that was obtained in the first step. The utilized CNN layers employed the Gaussian Error Linear Unit (GELU) function³⁹ as the activation function. Finally, to integrate the feature maps derived from F_r and F_i , the real and imaginary parts of the extracted feature maps, denoted by $\hat{F}_r$ and $\hat{F}_i$, respectively, were reconstructed into complex-valued representations as $F_{complex} = \hat{F}_r + i\hat{F}_i$. The complex norm of the $F_{complex}$ was then computed to obtain the integrated feature maps. Additionally, the dropout layer with a rate of 0.5, followed by a max pooling layer with a pool size of 4 and a stride of 2, was applied as the final step.

2. Raw EEG feature extraction module

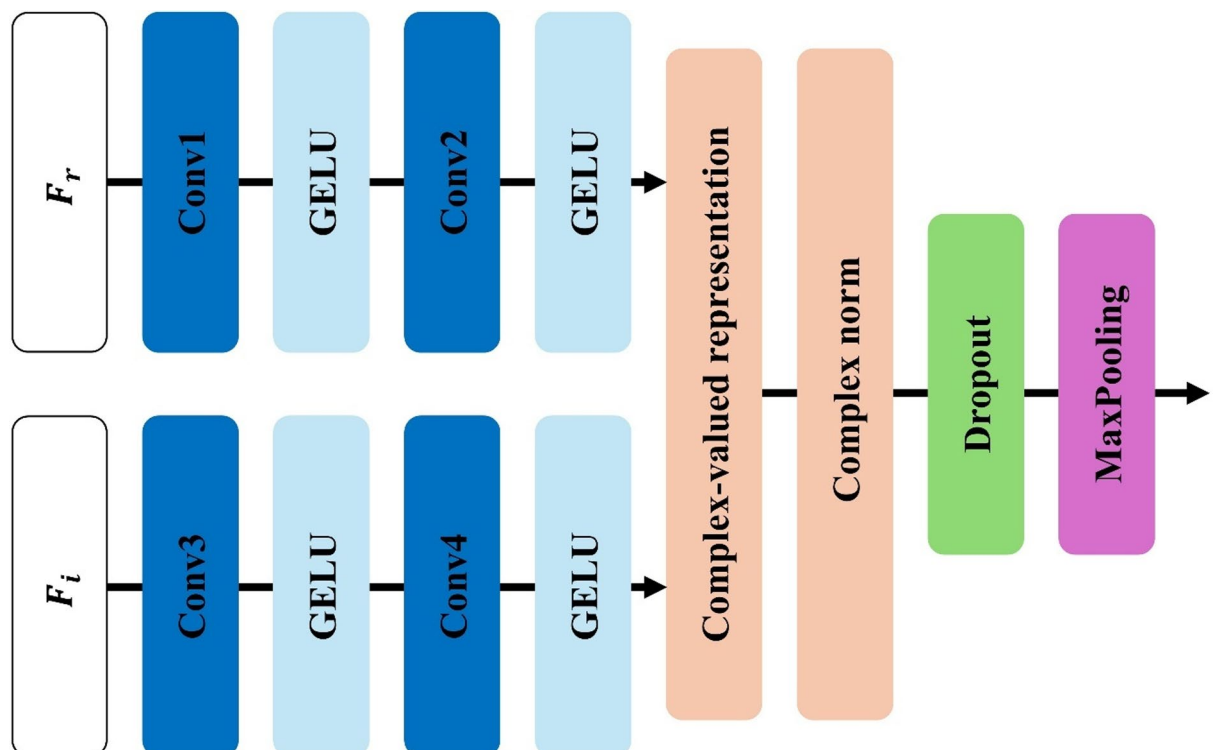


Fig. 2. Detailed architecture of PAC feature extraction module after concatenation of Hilbert-transformed low-frequency signal and high-frequency signal. F_r , features obtained by concatenating real part of Hilbert-transformed low-frequency signal and high-frequency signal; F_i , features obtained by concatenating imaginary part of Hilbert-transformed low-frequency signal and high-frequency signal; Conv, 2D convolutional neural networks; GELU, Gaussian Error Linear Unit.

It was assumed that PAC features would augment the capacity of the deep learning network to classify raw EEG signals. Therefore, the network architecture for the extraction of raw EEG features was not modified from conventional architectures. To extract these features, we employed a structure analogous to that utilized in various previous studies on ASSC, as illustrated in Fig. 3^{18,19,40,41}. This module is composed of two parallel blocks and a subsequent block: a CNN-based temporal feature extraction block with narrow kernels, a CNN-based frequency feature extraction block with wide kernels, and a refinement block. Each feature extraction block comprises identical components, including the first 1D CNN layer, followed by a max-pooling layer and a dropout layer with a rate of 0.5, the second 1D CNN layer, and the third 1D CNN layer, which is also followed by a max-pooling layer. These CNN layers used the GELU function as an activation function. The wide kernel size of the first 1D CNN layer was set to 200, considering the input signal's sampling rate of 100. Similarly, the narrow kernel size of the first 1D CNN layer was set to 50. Subsequently, the feature maps obtained from each module were concatenated along the temporal axis. To filter unnecessary information from the concatenated feature maps, refinement was conducted. In the refinement block, the concatenated feature maps were fed into a dropout layer with a rate of 0.5, followed by two additional 1D CNN layers to reduce the number of parameters. The activation function employed in these CNN layers was the Rectified Linear Unit (ReLU) function⁴². Finally, an average pooling layer was applied to reduce the sample size. All these structures and hyperparameters were optimized by repeated trial-and-error executions.

3. Classification module

To leverage both the information of PAC features and raw EEG features, the feature maps from the PAC feature extraction module and the raw EEG feature extraction module were concatenated along the temporal axis. For refinement, a 1D CNN layer was inserted, with a kernel size of 8 and a stride of 2, to extract feature maps that amalgamate information from both extraction modules. Subsequently, the integrated feature maps were flattened following their passage through a max pooling layer, with a pool size of 4 and a stride of 2, and were then subjected to a dropout layer with a rate of 0.5. After this process, two fully connected (FC) layers were employed to calculate the probabilities of EEG segments being classified into each sleep stage. The first FC layer employed the ReLU activation function, while the second FC layer utilized the softmax activation function, as shown in Fig. 1.

Weighted loss function

The imbalanced distribution of sleep stages is one of the most significant challenges in DL-based classification of sleep stages. To address this class imbalance issue, the weighted cross-entropy loss function developed by Eldele et al.⁴⁰ was employed, with weights adjusted according to the frequency of occurrence of each sleep stage. For example, higher weights were assigned to N1 stage with low prevalence, while lower weights were allocated to more prevalent N2 stage. Given that the one-hot encoded ground-truth label is represented as y_i^k and the predicted probability is denoted as $\hat{y}_i^k$, which indicates i -th sample for the stage k , where N is the batch size and K is the number of sleep stages, the weighted cross-entropy loss function, $Loss$, is defined by (2).

$$Loss = -\frac{1}{N} \sum_{k=1}^K \sum_{i=1}^N \omega_k y_i^k \log(\hat{y}_i^k) \quad (2)$$

$$\omega_k = \mu_k \cdot \max(1, \log(\mu_k N / N_k)), \quad (3)$$

Where ω_k stands for the weight of each stage, μ_k denotes the parameter for weight, and N_k is the number of samples in each stage. μ_k was assigned different values depending on the sleep stage as follows.

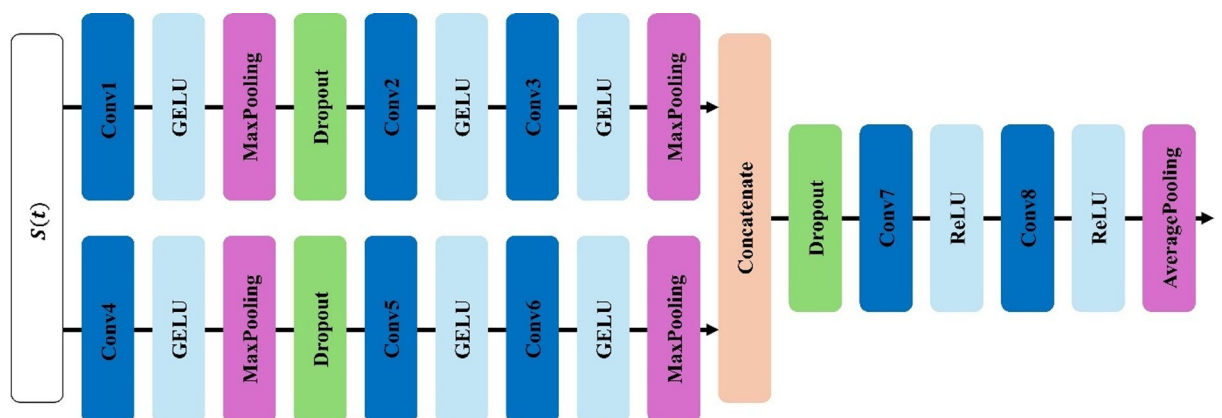


Fig. 3. Detailed architecture of Raw EEG feature extraction module. $S(t)$, raw EEG signal; Conv, 1D convolutional neural networks; GELU, Gaussian Error Linear Unit; ReLU, Rectified Linear Unit.

$$\mu_k = \begin{cases} a/K & k = W \\ b/K & k = N1 \\ c/K & k = N2 \\ d/K & k = N3 \\ e/K & k = REM \end{cases}, \quad (4)$$

Where a , b , c , d , and e are hyperparameters that are dependent on the distribution of the used dataset. They were set to 1.5, 1.8, 1, 1.5, and 1.2, respectively.

Model training

The leave-one-subject-out cross-validation method was employed to enhance the reliability of classification performance. To prevent overfitting, 24 participants were randomly divided into training and validation sets in a 21:3 ratio, excluding one designated for the test set. In this procedure, segments from the same participant were assigned to the same set to avoid information leakage between participants.

In the training phase, 100 epochs and a batch size of 128 were applied. To optimize the weighted loss function, AdamW⁴³ optimizer was employed considering its superior generalization capabilities compared to other optimizers. The parameters included 0.9 of β_1 , 0.999 of β_2 , and a weight decay of 1×10^{-6} . The cosine annealing learning rate scheduler⁴⁴ was employed to prevent the DL model from converging to a local minimum during the optimization process. The Initial learning rate was set to 1×10^{-3} , with a decay step of 1400. The parameter T_{mult} , M_{mult} , α were set to 1, 1, and 1×10^{-3} , respectively. All the training and classification processes of DL models were conducted using Python 3.9.12 version with NVIDIA Geforce RTX 4070 Ti Super.

Evaluation metrics

To evaluate the classification performance of each sleep stage, three per-class evaluation metrics, including Precision (PR), Recall (RE), and the per-class F1-score (F1) were used. The definitions of these metrics are as follows:

$$PR = \frac{TP}{TP+FP} \quad (5)$$

$$RE = \frac{TP}{TP+FN} \quad (6)$$

$$F1 = \frac{2 \times PR \times RE}{PR+RE} \quad (7)$$

Where TP , FP , and FN represent true positives, false positives, and false negatives, respectively. Additionally, the overall performance of sleep stage classification was evaluated using accuracy (ACC), macro-averaged F1-score ($MF1$), and Cohen's Kappa coefficient (κ)⁴⁵, which were calculated using the following equations:

$$ACC = \frac{\sum_{i=1}^K TP_i}{M} \quad (8)$$

$$MF1 = \frac{\sum_{i=1}^K F1_i}{K} \quad (9)$$

$$\kappa = \frac{ACC - P_C}{1 - P_C} \quad (10)$$

Where M is the total number of samples, P_C indicates the probability that the true and predicted sleep stages coincided, and K represents the number of sleep stages.

Models for comparison with SleepPACNet

To evaluate the classification performance of SleepPACNet objectively, we compared its performance with that of other DL models. Classification performances of the previous ASSC models, including SleepFCN¹⁸, DeepSleepNet⁴¹, and TinySleepNet²⁰ were implemented to prove that SleepPACNet outperformed these ASSC models. All the tested DL models were optimized so that they could achieve the highest ASSC accuracy. To ensure a fair and rigorous comparison with SleepPACNet, experimental data using the bipolar prefrontal EEG signals, model training from scratch, and evaluation were consistently applied to the previous ASSC models.

Ablation studies

To evaluate the contribution of the PAC feature extraction module in SleepPACNet, a conventional CNN-based DL model, denoted as SleepCNN, was implemented by excluding the PAC feature extraction module from SleepPACNet. Furthermore, several architectural variants were tested to justify the design choices of SleepPACNet from two perspectives: the efficacy of PAC feature extraction and hyperparameter optimality. First, regarding the efficacy of PAC extraction, the importance of explicitly exposing phase information was evaluated by assessing the case without the Hilbert transform. Within the same perspective, to demonstrate that the proposed module extracts more adaptive features, a case was conducted replacing the module with hand-crafted PAC features, specifically the MI calculated from the predefined frequency bands used in this study. Additionally, to investigate the most effective structural approach for complex-valued features, the performance was compared against a complex-valued CNN (CV-CNN) architecture, which was implemented as an alternative to the parallel CNN structure independently processing real and imaginary parts. Second, to ensure hyperparameter optimality, the influence of frequency band selection was examined by applying narrow frequency bands that precisely target

	Predicted					Per-class performance (%)		
	W	N1	N2	N3	REM	PR	RE	F1
W	2003	384	183	22	445	0.722	0.660	0.689
N1	210	484	358	3	450	0.270	0.322	0.293
N2	261	621	9510	584	903	0.871	0.801	0.834
N3	130	13	496	2805	70	0.815	0.798	0.807
REM	171	294	377	28	3857	0.674	0.816	0.738

Table 1. Confusion matrix and per-class performance of the SleepPACNet using Fp1–Fp2 signal.

Models	Overall performance			Per-class F1-score (%)				
	ACC (%)	MF1 (%)	κ	W	N1	N2	N3	REM
SleepPACNet (ours)	75.7	0.672	0.654	0.689	0.293	0.834	0.807	0.738
SleepFCN ¹⁸	55.2	0.496	0.413	0.542	0.158	0.611	0.663	0.507
DeepSleepNet ⁴¹	67.2	0.542	0.532	0.545	0.057	0.758	0.708	0.642
TinySleepNet ²⁰	71.8	0.566	0.575	0.607	0.012	0.800	0.740	0.672

Table 2. Comparison of the classification performance of SleepPACNet and other previous ASSC models using Fp1–Fp2 signal.

delta waves (0.5–4 Hz) and sleep spindles (12–15 Hz). Furthermore, the robustness of the model's regularization was evaluated by modifying the dropout rate to a lower value of 0.2. To ensure an objective evaluation, all hyperparameters and experimental settings were identically applied to SleepPACNet, except for the modified or excluded modules in each ablation case.

Statistical analysis

To examine which sleep stages showed significant improvement in classification performance, F1-scores of SleepPACNet and SleepCNN were compared for each sleep stage. The normality of the F1-score distributions was assessed by confirming that the absolute values of skewness and kurtosis were below 2 and 7, respectively⁴⁶. Since the F1-scores satisfied the normality assumption, a paired t-test was performed on the F1-scores for each sleep stage in SleepPACNet and SleepCNN. To address the multiple comparison issue, Bonferroni correction was applied by multiplying the probability by five, which corresponds to the number of sleep stages⁴⁷.

Results

Model classification performance

Table 1 presents the confusion matrix and per-class performance metrics of SleepPACNet. The N2 and N3 stages exhibited superior classification performance compared to other sleep stages, with F1-scores exceeding 0.8. In contrast, a large number of epochs in the N1 stage were misclassified into other sleep stages, with the exception of the N3 stage. Table 2 compares the classification performance of SleepPACNet with that of previous ASSC models. Both overall performance and per-class F1-scores were highest in SleepPACNet compared to all previous DL-based ASSC models, achieving a classification accuracy of 75.7%. Detailed per-subject classification accuracy is provided in the Supplementary Information to further demonstrate the generalizability and robustness of SleepPACNet (Supplementary Fig. 1).

Ablation studies of SleepPACNet

Table 3 summarizes the results of the ablation studies, where SleepPACNet serves as the reference model. As shown in the table, SleepPACNet outperformed SleepCNN in both overall performance and per-class F1-scores, improving classification accuracy by more than 2.4% compared to when the PAC feature extraction module was excluded. Regarding the efficacy of the PAC extraction module, replacing this module with hand-crafted PAC exhibited the lowest accuracy of 71.2%, which was even lower than that of SleepCNN. In contrast, while the case without the Hilbert transform and the use of CV-CNN architecture yielded higher accuracy than SleepCNN, both cases resulted in lower overall performance compared to SleepPACNet. Specifically, the case of using CV-CNN outperformed the case without the Hilbert transform in terms of overall performance. With respect to hyperparameter optimality, both the application of narrow frequency bands and the reduction of the dropout rate to 0.2 resulted in lower accuracy compared to original SleepPACNet.

Hypnogram

To illustrate the similarity between the ground-truth hypnogram and the predicted hypnogram, Fig. 4 displays the hypnograms of a participant whose sleep stages were predicted with the highest accuracy. The overall sleep structure of the predicted hypnogram closely resembles that of the true hypnogram; however, there are some parts in the predicted hypnogram where sleep stages transition rapidly analogous to noise. It is to be noted

Architectures	Overall performance			Per-class F1-score (%)				
	ACC (%)	MF1 (%)	κ	W	N1	N2	N3	REM
SleepPACNet	75.7	0.672	0.654	0.689	0.293	0.834	0.807	0.738
SleepCNN	73.3	0.646	0.623	0.636	0.265	0.831	0.795	0.705
Hand-crafted PAC	71.2	0.620	0.586	0.594	0.252	0.803	0.781	0.670
w/o Hilbert Transform	74.2	0.657	0.638	0.641	0.278	0.826	0.788	0.753
CV-CNN	75.1	0.666	0.646	0.664	0.296	0.836	0.793	0.739
Narrow frequency bands	74.3	0.658	0.639	0.680	0.268	0.833	0.798	0.710
Low Dropout (rate = 0.2)	72.6	0.641	0.614	0.628	0.272	0.818	0.787	0.702

Table 3. Ablation studies: Classification performance of SleepPACNet.

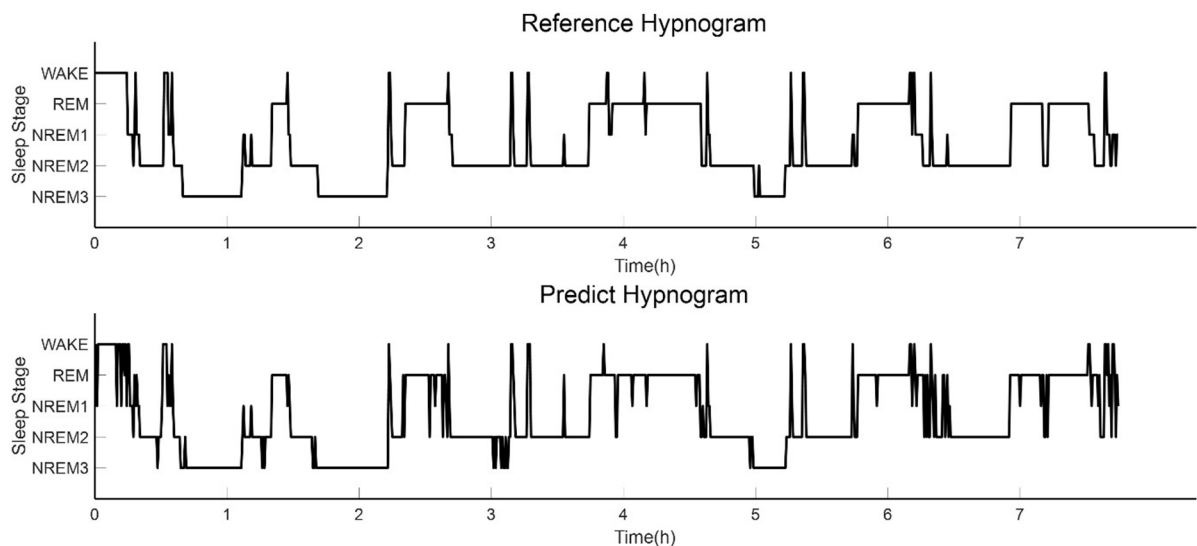


Fig. 4. The hypnogram of a subject who showed the best classification accuracy. The upper hypnogram was manually scored by an expert, while the lower hypnogram was automatically classified using SleepPACNet.

that the predicted hypnogram was obtained using a single-channel prefrontal EEG, whereas the ground-truth hypnogram was estimated using multimodal biomedical signals manually by a sleep expert.

Comparison of f1-score between SleepPACNet and SleepCNN

Figure 5 shows the F1-score distributions of SleepPACNet and SleepCNN across sleep stages. There is a significant difference in the F1-scores between SleepPACNet and SleepCNN for the REM stage (Bonferroni corrected $p = 0.002$; SleepPACNet: 0.729 ± 0.039 ; SleepCNN: 0.686 ± 0.041), with SleepPACNet achieving a significantly higher F1-score compared to SleepCNN. In constant, no significant differences were observed for other sleep stages, although SleepPACNet exhibited higher F1-scores than SleepCNN across all sleep stages.

Discussions

In this study, we developed SleepPACNet, a novel CNN-based model incorporating PAC information, and we evaluated the ASSC performance of SleepPACNet. Our findings indicate that SleepPACNet is capable of accurately classifying sleep stages utilizing bipolar prefrontal single-channel EEG, which is beneficial for capturing PAC information while maintaining superior classification performance. SleepPACNet demonstrated enhanced performance compared to other ASSC models, such as SleepCNN, SleepFCN, DeepSleepNet, and TinySleepNet. Although SleepPACNet did not show a statistically significant improvement in performance compared to SleepCNN during the N2 and N3 stages, it achieved robust classification performance with an F1-score exceeding 0.8, demonstrating its ability to accurately detect deep sleep. Additionally, a statistically significant improvement was observed during the REM stage, indicating that the extracted PAC features played a crucial role in distinguishing REM sleep patterns in a bipolar prefrontal single-channel EEG. While CNNs are proficient at extracting amplitude features from raw EEG signals, they face challenges in effectively capturing phase features. To address this issue, Kim et al.⁴⁸ introduced the HiRENet, which facilitates the extraction of phase features from raw EEG signals via the Hilbert transform, demonstrating an enhancement in classification performance within deep learning frameworks. To extract PAC features, the Hilbert transform was utilized to clarify the phase information associated with low-frequency EEG signals. The importance of PAC information

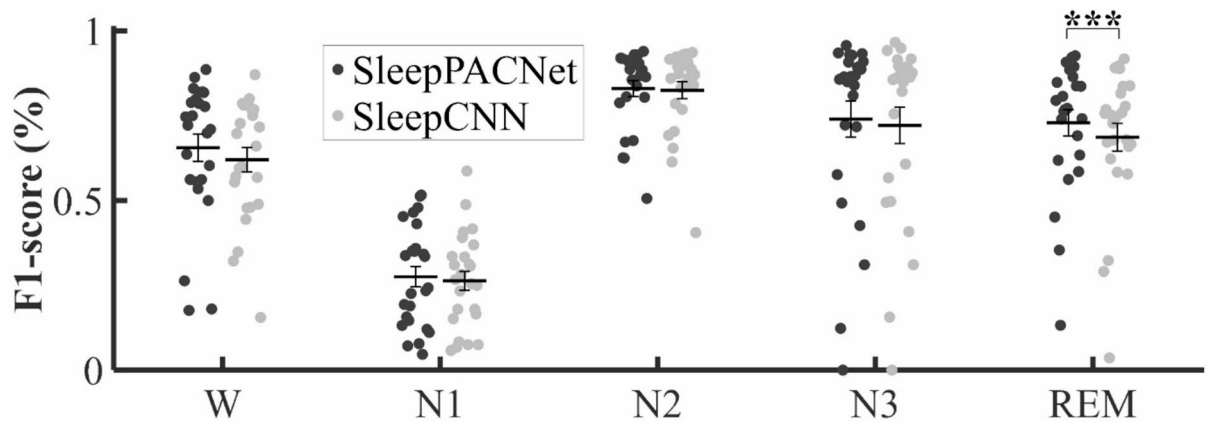


Fig. 5. The comparison of F1-scores across sleep stages between SleepPACNet and SleepCNN. The mean bars represent the average F1-scores for each sleep stage, while the error bars indicate the standard error of the mean. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

in the classification of sleep stages was evidenced by the superior performance of SleepPACNet in comparison to SleepCNN, the latter differing solely in its exclusion of the PAC feature extraction module present in SleepPACNet. This finding indicates that PAC features were effectively extracted by integrating the phase features of low-frequency EEG signals, obtained through the Hilbert transform, with the amplitude features of high-frequency EEG signals using CNN. These results imply that the PAC feature extraction module has significant potential to enhance the performance of sleep stage classification when incorporated into other models.

In particular, the results of the ablation studies provide strong evidence that the application of the Hilbert transform to reveal phase information for the automatic extraction of PAC features plays a critical role in enhancing performance. Among the ablation studies focused on the efficacy of the PAC extraction module, replacing this module with hand-crafted PAC resulted in the lowest performance. This indicates that the PAC extraction module enhances classification by automatically selecting the most suitable PAC features among various PAC metrics for sleep stage classification, rather than relying on predefined fixed metrics. Furthermore, the finding that the case without the Hilbert transform outperformed SleepCNN but remained inferior to SleepPACNet suggests that while the separation of frequency bands provides some performance improvement, the explicit exposure of phase information using the Hilbert transform makes a more substantial contribution to the classification performance. Finally, the slightly higher performance of SleepPACNet compared to the case using CV-CNN may reflect that our architecture is functionally sufficient to capture essential phase information. Consequently, the consistent superior performance of SleepPACNet across three ablation studies demonstrates that revealing phase information and automatically extracting PAC features is the optimal configuration for sleep stage classification.

Furthermore, the results of an ablation study on narrow frequency bands (e.g., 0.5–4 Hz for the low-frequency band and 12–15 Hz for the high-frequency band) exhibited lower classification performance compared with SleepPACNet, which utilizes 0.5–8 Hz for a low-frequency band and 8–30 Hz for a high-frequency band. The frequency bands of SleepPACNet were empirically selected to comprehensively capture the characteristic interactions of both NREM and REM sleep stages. The enhanced performance of SleepPACNet is attributed to the inclusion of broader frequency ranges, specifically a low-frequency band (0.5–8 Hz) that incorporates both the delta waves (0.5–4 Hz) prominent during NREM sleep and the theta waves (4–8 Hz), characteristic of REM sleep. Similarly, the high-frequency band (8–30 Hz) encompasses not only NREM sleep spindles (12–15 Hz) but also beta activity (15–30 Hz), which has been reported to interact with theta rhythms in the frontal lobe during REM sleep⁴⁴. This broad frequency band may enable the CNN to automatically extract cross-frequency relationships from both NREM and REM sleeps, such as delta–spindle coupling during NREM and theta–beta interactions during REM.

SleepPACNet also outperformed previous ASSC models in classifying the sleep stages using a bipolar prefrontal single-channel EEG signal. In our study, other previous ASSC models exhibited lower accuracy compared to the earlier research, which may be attributed to the use of a bipolar prefrontal single-channel EEG signal. Most prior studies employing single-channel EEG for ASSC used electrode pairs including central electrodes, such as Fpz–Cz, C3–M2, and C4–M1, which are recommended by the R&K and AASM manuals as key channels where sleep EEG features, such as sleep spindles and K-complex, are prominently represented^{7,8}. Accordingly, several studies utilizing Fp1–Fp2 signals have consistently reported lower performances; for instance, Shambroom et al.⁴¹ showed that ASSC using Fp1–Fp2 signal exhibited lower performance compared to that using PSG, and Stepnowsky et al.⁴² reported a classification accuracy of approximately 70% using Fp1–Fp2 signals. Consequently, the lack of distinct EEG characteristics in the prefrontal regions inevitably leads many state-of-the-art ASSC models to underperform in this study compared to their results in previous studies. However, the employment of a bipolar prefrontal single-channel EEG signal for ASSC is important from a practical perspective, such as when using a portable EEG device during sleep at home. This approach helps reduce noise caused by hair⁴⁹ and enables the use of electronic skin for EEG measurement. Therefore,

SleepPACNet has the potential to be effectively implemented in real-world applications for the classification of sleep stages by extracting both PAC information and other EEG characteristics from a bipolar prefrontal single-channel EEG signal.

Despite the practical advantages of using Fp1–Fp2 signals, the development of ASSC models is limited by the absence of an open database with bipolar prefrontal EEG signals. In this study, the Fp1–Fp2 signals were synthesized using Fp1–F3, Fp2–F4, and F3–F4 signals in DOD. However, synthesized signals may potentially differ from real signals in terms of noise characteristics. To verify the technical validity of this synthesis, an additional analysis was conducted by comparing the synthesized Fp1–Fp2 signals with real Fp1–Fp2 signals recorded during the eye-closed resting-state in an independent measurement. Detailed information regarding the measurement is provided in the Supplementary Information. The synthesized and real signals exhibited similar waveforms and showed high correlations in power spectral density ($r=0.910$) (Fig. 6). Furthermore, noise analysis using the FOOOF algorithm⁵⁰ demonstrated consistent 1/f noise profiles between the synthesized Fp1–Fp2 signals and the original Fp1–F3, Fp2–F4, and F3–F4 signals in DOD (Supplementary Fig. 2). While minor discrepancies may exist due to hardware factors such as electrode impedance, these results demonstrate that the synthesized Fp1–Fp2 signals effectively preserve the essential physiological characteristics of recorded Fp1–Fp2 signals. Nevertheless, the public release of large-scale, open-access Fp1–Fp2 datasets would be invaluable for further advancing practical tools that support sleep assessment and improvement.

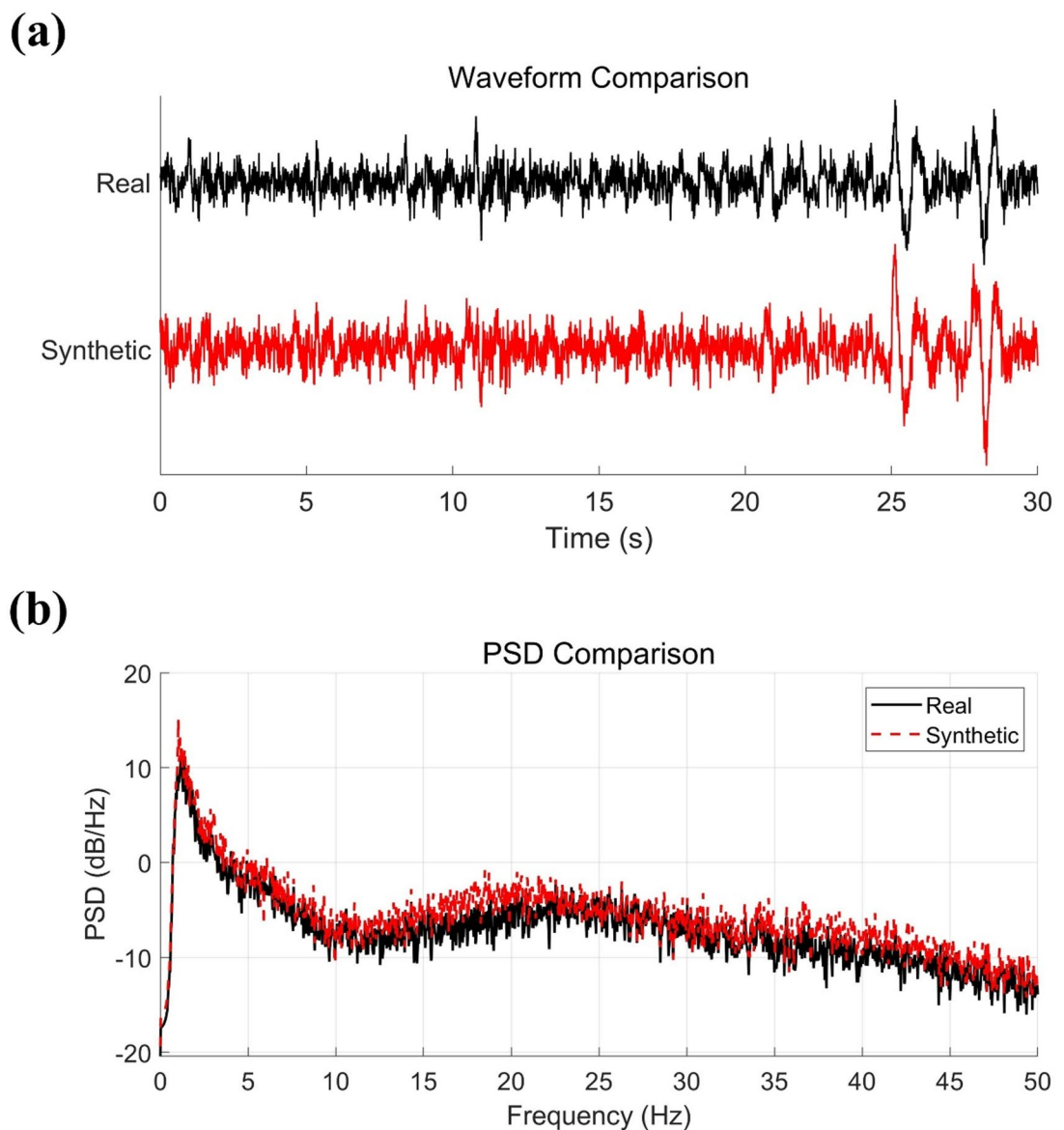


Fig. 6. Comparison between real and synthetic Fp1–Fp2 EEG signals. (a) The comparison of signal waveforms and (b) The comparison of PSD. The black lines represent the real EEG signal or features derived therefrom, while the red lines represent the synthetic EEG signal or its derived features. PSD: power spectral density.

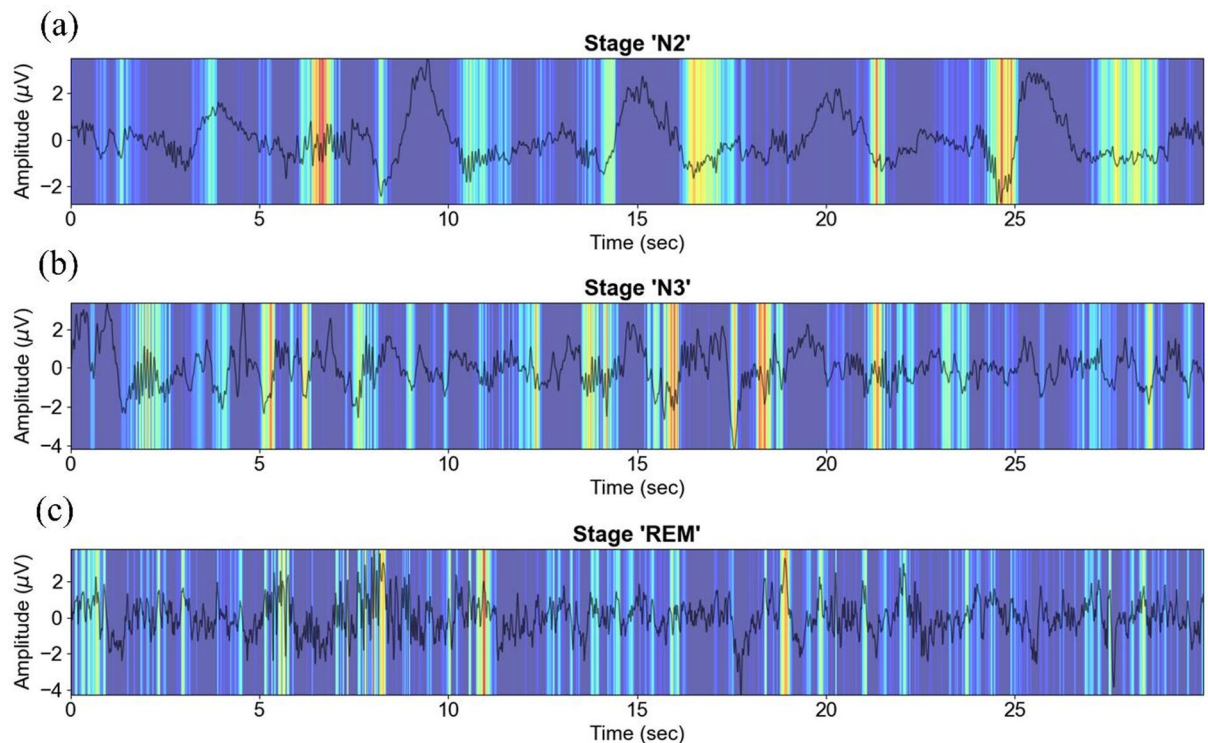


Fig. 7. Grad-CAM visualization of learned features. These graphs display the raw EEG signals overlaid with importance heatmaps (with color intensity representing the level of feature importance). (a) N2 and (b) N3: The model's attention concentrates on specific phases of delta waves, precisely where sleep spindle waveforms are clearly observed. (c) REM: The importance weights increase around the peaks of theta waves, coinciding with regions where beta activity is distinctly visible. These patterns confirm that the model automatically captures physiological delta-spindle coupling in NREM and theta-beta coupling in REM sleep.

Scored sleep stages are commonly used to assess sleep quality and facilitate its improvement. Among these stages, the N3 stage is recognized as one of the most important, as a deficiency in N3 stage can lead to fatigue, waking up feeling tired, and cognitive decline^{51,52}. Consequently, many studies have been conducted to increase slow oscillations of the N3 stage during sleep through auditory^{53,54} or electrical stimulation^{55,56}. However, indiscriminate stimulation can disrupt sleep rather than enhance it⁵⁴, underscoring the need for precise, closed-loop stimulation specifically targeting the N3 stage. To achieve this, an accurate ASSC model is essential for reliably identifying the N3 stage. SleepPACNet demonstrates strong potential in this regard, achieving an F1-score exceeding 0.8 for N3 stage detection. This capability marks a significant advancement in the commercialization of home-based stimulation systems, as discussed in relation to real-world applications.

Aside from sleep research, PAC features have been employed as important markers for traditional ML-based classification across various domains, including neuropsychiatry, motor imagery, and emotion recognition^{57–60}. Furthermore, many studies in these fields have attempted to use PAC features in DL-based classification. For example, Li et al.⁶¹ classified epileptic brain activity using handcrafted complex-valued PAC features with a DL-based approach utilizing a complex-valued CNN. However, handcrafted PAC features may not always represent the optimal PAC measure for each specific classification task, potentially limiting improvements in classification performance in previous studies. Therefore, SleepPACNet could provide an alternative approach to enhance classification performance in other fields, as it automatically extracts features.

Although the PAC features were successfully extracted using SleepPACNet, the precise role of these features in classifying sleep stages remains unclear due to the inherent nature of DLs' black box systems. This limitation emphasizes the necessity of investigating the extracted PAC features in greater detail to uncover the underlying neurophysiological phenomena associated with PAC during different sleep stages. Therefore, to verify whether the learned features align with physiological expectations using explainable artificial intelligence (XAI) methods, we visualized the class activation maps using Grad-CAM. As shown in Fig. 7, the heatmaps showed that the model's attention was concentrated at specific phases of delta waves, precisely where sleep spindle waveforms were clearly observed in NREM stages (N2 and N3). In the REM stage, the importance weights increased around the peaks of theta waves, coinciding with regions where beta activity was distinctly visible. These results confirm that SleepPACNet successfully learned the well-known delta-alpha coupling in NREM and theta-beta interaction in REM. Additionally, these XAI methods could further expand the application of PAC by enhancing our understanding of its mechanisms. For example, there have been some attempts to improve memory consolidation, which is a well-known phenomenon related to PAC that occurs when people fall asleep through the use of transcranial electrical stimulation (tES)^{33,62,63}. As more detailed insights into PAC parameters, such as

the timing of PAC synchronization, are obtained, tES may be employed to further enhance PAC synchronization and thereby strengthen memory consolidation⁶⁴.

There are several primary limitations in our study. Firstly, the absence of an open database containing bipolar prefrontal EEG signals with PAC information necessitated the theoretical generation of these signals to approximate those obtained through direct measurement. However, it is important to acknowledge that minor discrepancies may arise due to potential noise interference. Secondly, the limited number of sleep stage samples available constrains the application of deep learning-based classification methods. Thirdly, SleepPACNet was evaluated using only DOD, which may limit the generalization ability of our findings to different populations and recording setups. Consequently, further validation of SleepPACNet using additional datasets or diverse recording environments, with larger sample sizes obtained directly from bipolar prefrontal single-channel EEG signals, would significantly strengthen the evidence for the applicability of SleepPACNet. Additionally, although SleepPACNet primarily relies on CNNs to extract frequency-specific features, this architecture may have limitations in modeling long-range temporal dependencies across consecutive epochs. Previous ASSC studies have demonstrated that hybrid architectures combining CNNs with explicit temporal modeling modules can significantly enhance decoding performance. For instance, DeepSleepNet⁴¹ utilized Bidirectional LSTM (Bi-LSTM) to capture sequential patterns, and AttnSleepNet⁴⁰ employed a multi-head attention mechanism to learn temporal context from sleep EEG signals. Additionally, AtSiftNet⁶⁵ has also demonstrated that attention mechanisms can enhance EEG classification performance by effectively capturing temporal information. Therefore, a promising future direction is to integrate these temporal modeling mechanisms (e.g., Self-Attention or Bi-LSTM) into the SleepPACNet framework to better capture contextual information across sleep stages.

Conclusion

We developed a novel CNN-based ASSC model, SleepPACNet, which is capable of automatically extracting PAC features to classify sleep stages using single-channel EEG. The PAC features were extracted using CNN through the concatenation of a Hilbert-transformed low-frequency EEG signal, which enhances the accessibility of phase information, and a high-frequency EEG signal. To evaluate the classification performance of SleepPACNet, we used the bipolar prefrontal single-channel EEG signal, Fp1–Fp2, as it provides valuable PAC information and offers high potential for practical applications. The classification performance of SleepPACNet outperformed that of SleepCNN, indicating that the automatically extracted PAC features from SleepPACNet contribute to more accurate sleep stage classification.

However, the small sample size, the limited generalization arising from using a single dataset, and the use of generated signals, which may be affected by potential noise issues, may constrain our study. Future work should focus on improving classification performance and generalization by directly measuring bipolar prefrontal single-channel EEG signals from a larger number of participants across diverse populations or recording setups during sleep. Furthermore, the practical applicability of SleepPACNet can be further validated by employing portable EEG devices in subsequent studies. Additionally, the implementation of XAI methods in further studies could provide insights into the role of PAC features as biomarkers in classification, thereby broaden our understanding of the sleep phenomena associated with PAC.

Data availability

The Dreem Open datasets are publicly available and can be accessed at [<https://zenodo.org/records/15900394>] (<https://zenodo.org/records/15900394>). Further detailed information regarding the dataset is provided in the associated publication ([<https://doi.org/10.1109/TNSRE.2020.3011181>] (<https://doi.org/10.1109/TNSRE.2020.3011181>)).

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Author contributions

This study was designed by T. Lee, J. Shon, and C. Im. Data selection and preprocessing were performed by T. Lee and H. Nam. Model development, training, and evaluation were conducted by T. Lee, J. Shon and C. Im supervised the entire project. The manuscript was written by T. Lee, J. Shon, and C. Im. All authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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